Weak Direct-Link Extension for Active STAR-RIS Assisted UAV Networks With NOMA

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In practical scenarios, assuming a completely blocked direct link between the UAV and the ground users may be restrictive, since realistic wireless environments may still allow a weak or partially obstructed direct link [24]. Therefore, in this supplementary note, we extend the considered system to a hybrid scenario where a weak direct "UAV to user" link coexists with the "STAR-RIS assisted" link. In fact, the proposed model in our work can be easily extended to the scenario including direct links.

I. Extended Signal Model and SINR Expressions

Considering the influence of the introduced direct "UAV to user" link, the received signal at $\mathcal{U}_\kappa$ can be rewritten as

$$Y_\kappa = \rho\left(\mathbf{h}\boldsymbol{\Theta}_\kappa \mathbf{g}_\kappa \Phi_\kappa X + \boldsymbol{\Theta}_\kappa \mathbf{g}_\kappa \mathbf{n}_1\right) + h_{d,\kappa}\Phi_{d,\kappa} X + n_0 \quad (1)$$

where $h_{d,\kappa} = \varrho \hat{h}_{d,\kappa} + \bar{\varrho}\Delta h_{d,\kappa}$ denotes the channel coefficient of the direct "UAV to user" link and $\Phi_{d,\kappa} = d_{u\kappa}^{-0.5o_3}$ is the path loss of the direct link, in which $d_{u\kappa}$ is the distance between UAV and $\mathcal{U}_\kappa$, o_3 represents the corresponding path loss exponent.

It is worth noting that, when the direct link is taken into account, the STAR-RIS phase shifts are configured to ensure that the cascaded signals are aligned with the direct-link phase $\phi_{d,\kappa}$ [37]. Therefore, the phase error $\Delta\phi_n$ is rewritten as

$$\Delta\phi_n = \varphi_n^\kappa - \theta_n - \xi_n + \phi_{d\kappa} \quad (2)$$

where $\Delta\phi_n$ still follows a Generalized Uniform distribution. In addition, it should be reminded that the introduction of direct-link phase only changes the phase reference for alignment, while the residual phase error is still caused by the discrete phase quantization of STAR-RIS.

Further extending **Eq.1**, SINR expressions of the hybrid active STAR-RIS assisted UAV network considering the influence of direct link can be derived as

$$\gamma_s = \frac{(1-a)P|\chi_2|^2}{aP|\chi_2|^2 + \rho^2\sigma_1^2 d_{s2}^{-o_2}|G_2|^2 + P\sigma_e^2 + \sigma_0^2} \quad (3)$$

$$\gamma_2 = \frac{aP|\chi_2|^2}{\rho^2\sigma_1^2 d_{s2}^{-o_2}|G_2|^2 + \beta_{RI}\mathbb{A} + \left(1+\beta_{RI}\right)aP\sigma_e^2 + \sigma_0^2} \quad (4)$$

$$\gamma_1 = \frac{(1-a)P|\chi_1|^2}{aP|\chi_1|^2 + \rho^2\sigma_1^2 d_{s1}^{-o_2}|G_1|^2 + P\sigma_e^2 + \sigma_0^2} \quad (5)$$

where $\chi_\kappa = \rho\mathbf{h}\boldsymbol{\Theta}_\kappa\mathbf{g}_\kappa\Phi_\kappa + h_{d,\kappa}\Phi_{d,\kappa}$ and $\mathbb{A} = (1-a)P|\chi_2|^2$.

II. Approximate Analytical Treatment

To calculate the analytical expressions of downlink metrics for hybrid active STAR-RIS assisted UAV network considering the influence of direct link, we just need to re-define $|\chi_\kappa|^2$, then follow the derivation method provided in our work. Taking γ_s as an example, $F_{\gamma_s}(\mathcal{X})$ is recomputed as **Eq.6** (shown at the top of the next page)

After that, we should quantitatively confirm the influence of the distance $d_{u\kappa}$ of direct "UAV to user" link (taking d_{u2} as the example). Denoting the coordinates of UAV and $\mathcal{U}_2$ by (z, θ_A, h) and $(y_2, \theta_B, 0)$, respectively, the corresponding distance d_{u2} can be expressed as

$$d_{u2} = \sqrt{z^2 + y_2^2 - 2zy_2\cos(\theta_A - \theta_B) + h^2} \quad (7)$$

According to the constructed CRWP and SRWP mobility models in our work, the PDFs of UAV and $\mathcal{U}_2$ are respectively given by

$$\begin{cases} f_A(z,\theta_A,h) = \frac{1}{2\pi} f_{z_{us}}(z) f_H(h); \\ f_B(y_2,\theta_B) = \frac{1}{2\pi} f_{d_{s2}}(y_2) \end{cases} \quad (8)$$

To avoid high-dimensional integral computation, we introduce the following approximation, i.e., replacing d_{u2} with $\mathrm{E}[d_{u2}]$ (this method has been adopted in existing studies, such as **Ref.[1]** and **Ref.[2]**), which can be computed as

$$\mathrm{E}[d_{u2}] = \int_0^{R_1}\int_0^{R_2}\int_0^{H}\int_0^{2\pi}\int_0^{2\pi} d_{u2} f_A(z,\theta_A,h)\cdot f_B(y_2,\theta_B)\; d\theta_A d\theta_B dh dy_2 dz \quad (9)$$

Since substituting **Eq.9** into $F_{\gamma_s}(\mathcal{X})$ would result in complex computation, we introduce the integral identity from Gradshteyn and Ryzhik [31] to guarantee tractability, i.e.,

$$\frac{1}{2\pi}\int_0^{2\pi}\sqrt{u - v\cos\Delta\theta}\, d\Delta\theta = \frac{2}{\pi}\sqrt{u+v}\,E\Big(\sqrt{\frac{2v}{u+v}}\Big) \quad (10)$$

where $E(x) = \int_0^{\frac{\pi}{2}}\sqrt{1 - x^2\sin^2\theta}\,d\theta$ denotes the complete elliptic integral of the second kind [31].

Replacing u, v and $\Delta\theta$ in **Eq.10** with $z^2 + y_2^2 + h^2$, $2zy_2$ and $\theta_A - \theta_B$, respectively, then $\mathrm{E}[d_{u2}]$ is accordingly rewritten

$$
\begin{aligned}
F_{\gamma_s}(\mathcal{X}) &= \mathrm{Prb}\{\gamma_s < \mathcal{X}\} = \mathrm{E}\left[F_{|\chi_2|^2}\left(\frac{\mathcal{X}(\rho^2\sigma_1^2|G_2|^2 y_2^{-o_2} + P\sigma_e^2 + \sigma_0^2)}{P[1-(\mathcal{X}+1)a]}\middle|\Phi_2, \Phi_{d,2}, G_2\right)\right] = \\
&\int_0^{R_1}\int_0^{R_2}\int_0^{+\infty} F_{|\chi_2|^2}\left(\frac{\mathcal{X}(\rho^2\sigma_1^2|G_2|^2 y_2^{-o_2} + P\sigma_e^2 + \sigma_0^2)}{P[1-(\mathcal{X}+1)a]}\middle|\Phi_{d,2}\right) f_{d_{u2}}(x, y_2) f_g(G_2)\, dG_2 dy_2 dx
\end{aligned}
\tag{6}
$$

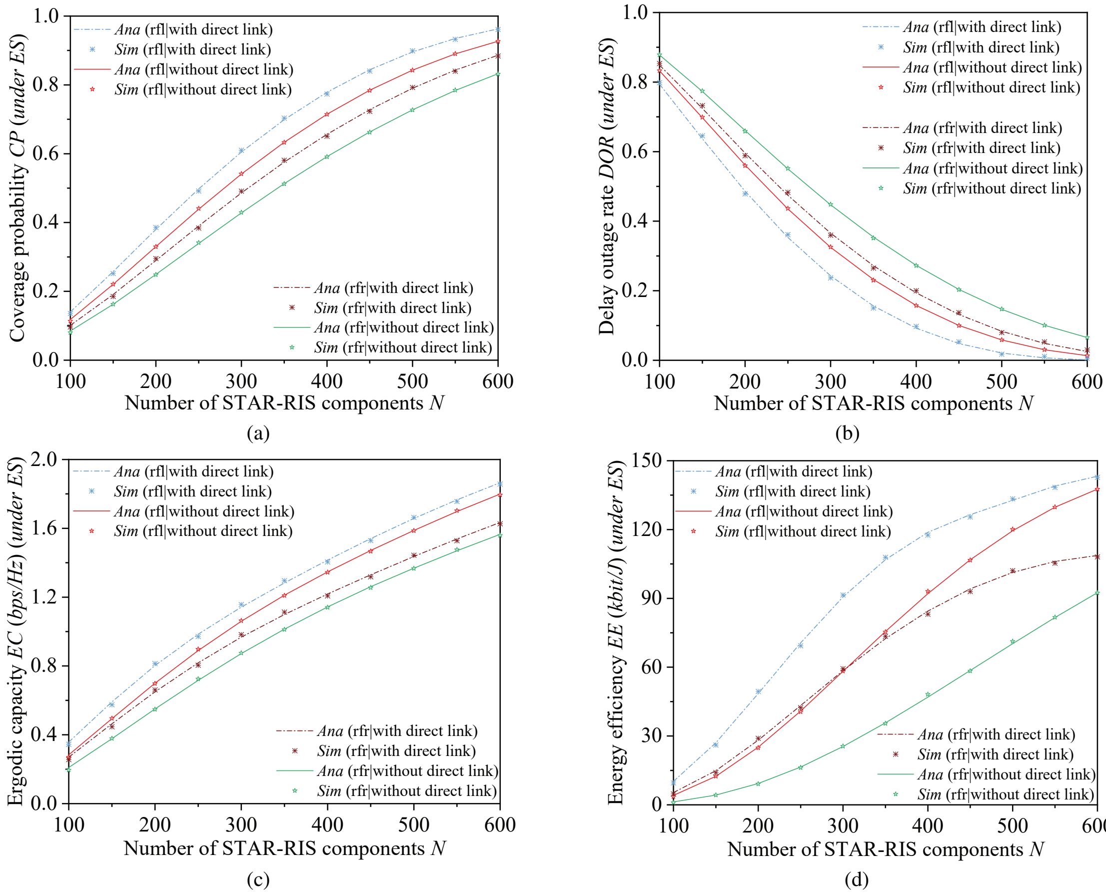

Fig. 1. Our work versus hybrid scenario with direct link under ES protocol.

as a tractable integral form, i.e.,

$$
\begin{aligned}
\mathrm{E}[d_{u2}] &= 4\pi \int_0^{R_1}\int_0^{R_2}\int_0^{H} f_A(z, \theta_A, h) f_B(y_2, \theta_B)\cdot \\
&\left[\frac{1}{2\pi}\int_0^{2\pi} d_{u2} d\Delta\theta\right] dh dy_2 dz \\
&= 8\int_0^{R_1}\int_0^{R_2}\int_0^{H} f_A(z, \theta_A, h) f_B(y_2, \theta_B)\cdot \\
&\sqrt{(z+y_2)^2+h^2}\, E\left(\sqrt{\frac{4zy_2}{(z+y_2)^2+h^2}}\right) dh dy_2 dz
\end{aligned}
\tag{11}
$$

Further applying Gaussian-Chebyshev quadrature method, the value of $\mathrm{E}[d_{u2}]$ can be obtained. Substituting $\mathrm{E}[d_{u2}]$ into $F_{\gamma_s}(\mathcal{X})$, the approximate expressions of downlink metrics of the system involving the direct link can be accordingly derived. Since the remaining derivation is similar to that already presented in the revised paper, the detailed procedure is not repeated here.

III. Performance Evaluation

Using the same parameters setting of the second experiment group (Section III.B) and set $o_3 = 2.6$, **fig. 1 and fig. 2 exhibit the influence of number of active STAR-RIS components N on the downlink performance of hybrid active STAR-RIS assisted UAV networks considering the influence of direct link under ES and MS protocols, respectively**.

It is easy to find that with the increase in the number of active STAR-RIS components, the downlink performance of hybrid active STAR-RIS assisted UAV network considering the influence of direct link is also improved. The reason is simple, i.e., increasing the number of active STAR-RIS components enhances the power gain $|\chi_\kappa|^2$. Similarly, $\mathcal{U}_2$ attains better downlink performance than $\mathcal{U}_1$ and executing ES protocol yields superior performance compared to executing MS protocol (analyzed in our article). Furthermore, in contrast to the scenario where no direct "UAV to user" link exists, it is clear that introducing the direct link consistently improves system performance across all metrics. In particular, due to the

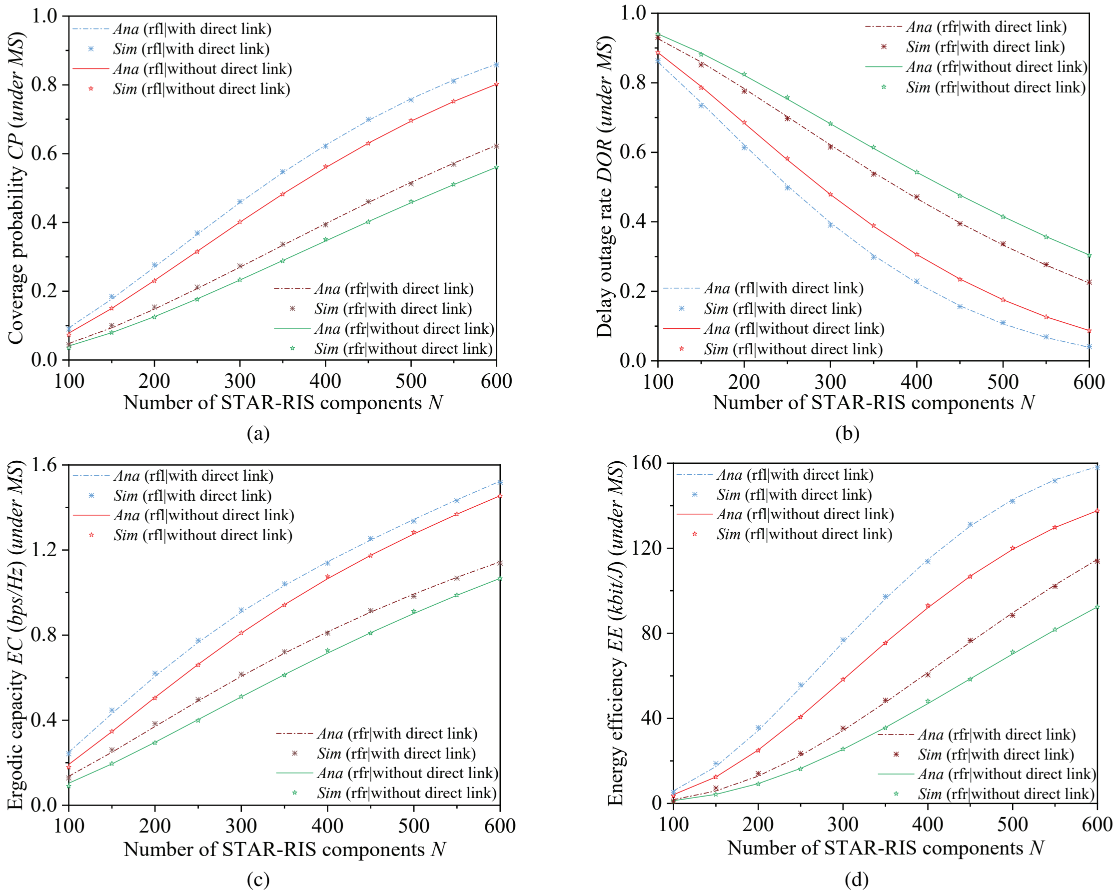

Fig. 2. Our work versus hybrid scenario with direct link under MS protocol.

introduced direct link, the coverage probability CP, ergodic capacity EC and energy efficiency EE become higher, while the delay outage rate DOR becomes lower. This is because the received signal now consists of both the STAR-RIS assisted path and the direct transmission path, which effectively provides additional power gain.

Through comparing the simulation results with numerical analysis results, it can be drawn that the proposed model of our work is still accurate for hybrid active STAR-RIS assisted UAV networks considering the influence of direct link. In addition, from the above derivation, it is easy to find that the proposed model has strong scalability for more complicated downlink transmission scenarios.

Ref.[1] and **Ref.[2]** are listed as follows:

Ref.[1]: D. Jia, Y. Zhong, X. Zhou and X. Ge, “Energy Efficiency Enhancement in RIS-Assisted Networks: Deployment and Phase Shifter Configuration,” *IEEE Trans. Green Commun. Netw.*, vol. 9, no. 3, pp. 1122–1137, Sep. 2025.

Ref.[2]: L. Wei, K. Wang, C. Pan and M. Elkashlan, “Average Error Probability for UAV-RIS Enabled Short Packet Communications,” *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2912–2917, Feb. 2024.